

Digital Payment Fraud Detection

SQL Business Case Study

BUSINESS CASE STUDY

Digital Payment
Fraud Detection

A SQL-Based Exploratory Analysis for a Financial Services Business

Dataset: 7,500 Transactions | 15 Features | 6.52% Fraud Rate

Tools: PostgreSQL | Aggregations | GROUP BY | CASE WHEN | Basic Window Functions

1. Project Overview

As a business stakeholder in a digital payments and financial services company, understanding and controlling fraud is mission-critical. This case study simulates a real-world scenario where a Data Analyst is tasked by the business to explore transaction data, identify simple fraud patterns, and present clear, data-backed insights using SQL.

1.1 Business Context

Our platform processes thousands of digital payment transactions daily across UPI, Cards, NetBanking, and Wallets. Even a small percentage of fraudulent transactions translates into significant financial losses and customer trust erosion. The goal of this analysis is to answer practical, stakeholder-relevant questions using foundational SQL skills and present a fraud insights summary that the Risk and Operations teams can act on.

1.2 Dataset Summary

Column Name	Data Type / Range	Business Meaning
transaction_id	String (T1–T7500)	Unique identifier per transaction
user_id	String (U#####)	Unique customer identifier
transaction_amount	Float ₹50.58 – ₹49,985.90	Value of the transaction in INR

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transaction_type	Payment / Withdrawal / Transfer	Nature of the transaction
payment_mode	Card / UPI / NetBanking / Wallet	Payment channel used
device_type	Web / iOS / Android	Device used for transaction
device_location	Delhi / Mumbai / Bangalore / Chennai / Hyderabad	City where device is located
account_age_days	Integer 10 – 1,999 days	How old the user account is
transaction_hour	Integer 0 – 23 (24-hr)	Hour of the day transaction occurred
previous_failed_attempts	Integer 0 – 4	Prior failed transaction attempts by user
avg_transaction_amount	Float User's historical average (INR)	Baseline for spotting unusual amounts
is_international	0 / 1 (Binary flag)	Whether the transaction is cross-border
ip_risk_score	Float 0.0 – 1.0	Risk score from IP intelligence (higher = riskier)
login_attempts_last_24h	Integer 1 – 9	Login activity in past 24 hours
fraud_label	0 = Legitimate 1 = Fraud	Target variable — ground truth fraud flag

2. Key Business Metrics at a Glance

TOTAL TRANSACTIONS 7,500 <i>Full dataset analysed</i>	FRAUD TRANSACTIONS 489 <i>6.52% fraud rate</i>	AVG TRANSACTION ₹24,813 <i>Across all transactions</i>	INTL TRANSACTIONS 755 <i>10.07% of all transactions</i>
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3. Stakeholder Business Questions

The following 12 questions are framed from the perspective of a Business Owner / Stakeholder at a digital payments company.

Category A — Fraud Overview

Q1 *[Executive Overview]*

What percentage of our total transactions are fraudulent, and what is their total value?

Q2 *[Amount Analysis]*

Is there a noticeable difference between the average transaction amount for fraud vs. legitimate transactions?

Category B — Payment Channel & Transaction Type Risk

Q3 *[Channel Risk]*

Which payment mode (UPI / Card / NetBanking / Wallet) has the highest fraud rate?

Q4 *[Transaction Risk]*

Which transaction type (Payment / Withdrawal / Transfer) sees the most fraud?

Category C — Behavioural & Account Risk Signals

Q5 *[Behaviour Risk]*

Do users with more previous failed attempts show a higher fraud rate?

Q6 *[Account Risk]*

Are newer accounts (low account_age_days) more prone to fraud than older accounts?

Q7 *[Behavioural Signal]*

Does a high number of login attempts in the last 24 hours correlate with fraud?

Q8 [IP Intelligence]**Is there a relationship between IP risk score and the likelihood of fraud?**

Category D — Geographic & Device Analysis

Q9 [Geographic Risk]**Which city has the highest number of fraudulent transactions?****Q10** [Device Risk]**Which device type (Web / iOS / Android) is most associated with fraud?****Q11** [International Risk]**Are international transactions more prone to fraud compared to domestic ones?**

Category E — Time-Based Patterns

Q12 [Temporal Risk]**At what hours of the day do most fraud transactions happen?**

4. SQL Queries with Business Answers

The following queries are written for PostgreSQL. The table is assumed to be named `fraud_transactions`. Each query uses simple, interview-friendly SQL — aggregate functions, GROUP BY, and CASE WHEN — to keep the logic easy to explain and defend in an interview.

Q1 — Overall Fraud Rate & Value at Risk

Business Context

As a stakeholder, the first thing I want to know is: how big is our fraud problem, in both transaction count and money terms?

```
SELECT
    COUNT(*) AS
total_transactions,
    SUM(CASE WHEN fraud_label = 1 THEN 1 ELSE 0 END) AS fraud_transactions,
    ROUND(
        SUM(CASE WHEN fraud_label = 1 THEN 1.0 ELSE 0 END)
        / COUNT(*) * 100, 2
    ) AS fraud_rate_pct,
    ROUND(SUM(CASE WHEN fraud_label = 1
        THEN transaction_amount ELSE 0 END)::NUMERIC, 2) AS value_at_risk_inr
FROM digital_payment_fraud;
```

Business Answer:

- 489 out of 7,500 transactions are fraudulent — a 6.52% fraud rate.
- Total value at risk from fraudulent transactions is roughly ₹1.18 Crore.

Q2 — Average Amount: Fraud vs. Legitimate

```
SELECT
    fraud_label,
    COUNT(*) AS transaction_count,
    ROUND(AVG(transaction_amount)::NUMERIC, 2) AS avg_amount_inr,
    ROUND(MIN(transaction_amount)::NUMERIC, 2) AS min_amount_inr,
    ROUND(MAX(transaction_amount)::NUMERIC, 2) AS max_amount_inr
FROM digital_payment_fraud
GROUP BY fraud_label
ORDER BY fraud_label;
```

Business Answer:

- Average fraud transaction (₹24,256) is actually very close to the average legitimate one (₹24,852).
- This tells us fraudsters aren't only targeting high-value transactions — amount alone isn't a reliable fraud signal.

Q3 — Fraud Rate by Payment Mode

Business Context

The Product team wants to know which payment channel has the weakest fraud controls so resources can be allocated to improve that channel.

```
SELECT
    payment_mode,
    COUNT(*)                                AS
total_transactions,
    SUM(CASE WHEN fraud_label = 1 THEN 1 ELSE 0 END) AS fraud_count,
    ROUND(
        SUM(CASE WHEN fraud_label = 1 THEN 1.0 ELSE 0 END)
        / COUNT(*) * 100, 2
    )                                       AS fraud_rate_pct
FROM digital_payment_fraud
GROUP BY payment_mode
ORDER BY fraud_rate_pct DESC;
```

Business Answer:

- Card payments have the highest fraud count (135 cases) among all channels.
- Action: Strengthen authentication (OTP / 3D Secure) specifically for Card transactions.

Q4 — Fraud by Transaction Type

```
SELECT
    transaction_type,
    COUNT(*)                                AS total_txns,
    SUM(CASE WHEN fraud_label = 1 THEN 1 ELSE 0 END) AS fraud_count,
    ROUND(
        SUM(CASE WHEN fraud_label = 1 THEN 1.0 ELSE 0 END)
        / COUNT(*) * 100, 2
    )                                       AS fraud_rate_pct
FROM digital_payment_fraud
GROUP BY transaction_type
ORDER BY fraud_rate_pct DESC;
```

Business Answer:

- Withdrawals see the most fraud (170 cases), followed closely by Transfers (160 cases).
- Action: Add an extra confirmation step for Withdrawal and Transfer requests above a set amount.

Q5 — Failed Attempts vs. Fraud Rate

Business Context

Does a history of failed attempts actually predict fraud? This helps the Risk team decide when to flag an account for review.

```
SELECT
    previous_failed_attempts,
    COUNT(*)                                AS total_txns,
    SUM(CASE WHEN fraud_label = 1 THEN 1 ELSE 0 END) AS fraud_count,
    ROUND(
        SUM(CASE WHEN fraud_label = 1 THEN 1.0 ELSE 0 END)
        / COUNT(*) * 100, 2
    )                                        AS fraud_rate_pct
FROM digital_payment_fraud
GROUP BY previous_failed_attempts
ORDER BY previous_failed_attempts;
```

Business Answer:

- Fraud rate rises gradually as previous_failed_attempts increases.
- Action: Consider mandatory re-verification once an account crosses 3 failed attempts.

Q6 — Account Age vs. Fraud Risk

```
SELECT
    CASE
        WHEN account_age_days < 30 THEN 'New (< 30 days)'
        WHEN account_age_days < 180 THEN 'Recent (30-180 days)'
        WHEN account_age_days < 365 THEN 'Established (180-365 days)'
        WHEN account_age_days < 730 THEN 'Mature (1-2 years)'
        ELSE 'Loyal (> 2 years)'
    END AS
    account_age_segment,
    COUNT(*)                                AS total_txns,
    SUM(CASE WHEN fraud_label = 1 THEN 1 ELSE 0 END) AS fraud_count,
    ROUND(
        SUM(CASE WHEN fraud_label = 1 THEN 1.0 ELSE 0 END)
        / COUNT(*) * 100, 2
    )                                        AS fraud_rate_pct
FROM digital_payment_fraud
GROUP BY account_age_segment
ORDER BY MIN(account_age_days);
```

Business Answer:

- Surprisingly, fraud is fairly evenly spread across account age groups — new accounts aren't dramatically riskier.
- This is a useful, honest finding: not every assumption about 'new account = risky' holds true in this data.

Q7 — Login Attempts as a Fraud Signal

```

SELECT
    login_attempts_last_24h,
    COUNT(*)                                AS total_txns,
    SUM(CASE WHEN fraud_label = 1 THEN 1 ELSE 0 END) AS fraud_count,
    ROUND(
        SUM(CASE WHEN fraud_label = 1 THEN 1.0 ELSE 0 END)
        / COUNT(*) * 100, 2
    )                                         AS fraud_rate_pct
FROM digital_payment_fraud
GROUP BY login_attempts_last_24h
ORDER BY login_attempts_last_24h;

```

Business Answer:

- Fraud rate trends slightly higher among users with more login attempts in the last 24 hours.
- Action: Flag accounts with unusually high login activity (7+ attempts) for extra verification.

Q8 — IP Risk Score vs. Fraud

```

SELECT
    CASE
        WHEN ip_risk_score < 0.2 THEN 'Very Low (0.0-0.2)'
        WHEN ip_risk_score < 0.4 THEN 'Low (0.2-0.4)'
        WHEN ip_risk_score < 0.6 THEN 'Medium (0.4-0.6)'
        WHEN ip_risk_score < 0.8 THEN 'High (0.6-0.8)'
        ELSE 'Very High (0.8-1.0)'
    END                                     AS ip_risk_bucket,
    COUNT(*)                                AS total_txns,
    SUM(CASE WHEN fraud_label = 1 THEN 1 ELSE 0 END) AS fraud_count,
    ROUND(
        SUM(CASE WHEN fraud_label = 1 THEN 1.0 ELSE 0 END)
        / COUNT(*) * 100, 2
    )                                         AS fraud_rate_pct
FROM digital_payment_fraud
GROUP BY ip_risk_bucket
ORDER BY MIN(ip_risk_score);

```

Business Answer:

- Fraud rate does not increase sharply with ip_risk_score alone in this dataset — it works better combined with other signals than on its own.

Q9 — Fraud by City

```

SELECT
    device_location,
    COUNT(*)                                AS total_txns,
    SUM(CASE WHEN fraud_label = 1 THEN 1 ELSE 0 END) AS fraud_count,
    ROUND(
        SUM(CASE WHEN fraud_label = 1 THEN 1.0 ELSE 0 END)

```



```
        / COUNT(*) * 100, 2
    )
    AS fraud_rate_pct
FROM digital_payment_fraud
GROUP BY device_location
ORDER BY fraud_count DESC;
```

Business Answer:

- Hyderabad records the highest fraud volume (115 cases); Delhi the lowest (83 cases).
- Action: Consider city-level monitoring thresholds, starting with Hyderabad and Mumbai.

Q10 — Fraud by Device Type

```
SELECT
    device_type,
    COUNT(*)
    AS total_txns,
    SUM(CASE WHEN fraud_label = 1 THEN 1 ELSE 0 END)
    AS fraud_count,
    ROUND(
        SUM(CASE WHEN fraud_label = 1 THEN 1.0 ELSE 0 END)
        / COUNT(*) * 100, 2
    )
    AS fraud_rate_pct
FROM digital_payment_fraud
GROUP BY device_type
ORDER BY fraud_rate_pct DESC;
```

Business Answer:

- Android devices show a slightly higher fraud rate compared to iOS and Web.

Q11 — International vs. Domestic Fraud

```
SELECT
    CASE WHEN is_international = 1
        THEN 'International' ELSE 'Domestic'
    END
    AS transaction_scope,
    COUNT(*)
    AS total_txns,
    SUM(CASE WHEN fraud_label = 1 THEN 1 ELSE 0 END)
    AS fraud_count,
    ROUND(
        SUM(CASE WHEN fraud_label = 1 THEN 1.0 ELSE 0 END)
        / COUNT(*) * 100, 2
    )
    AS fraud_rate_pct
FROM digital_payment_fraud
GROUP BY transaction_scope;
```

Business Answer:

- Domestic transactions actually have a marginally higher fraud rate (6.64%) than international (5.43%).
- This is a great talking point in an interview — it challenges the common assumption that 'cross-border = riskier.'

Q12 — Fraud by Hour of Day

Business Context

Our Operations team wants to know if there's a specific time window when fraud spikes, so they can staff up monitoring during those hours.

```
SELECT
    transaction_hour,
    COUNT(*)                                AS total_txns,
    SUM(CASE WHEN fraud_label = 1 THEN 1 ELSE 0 END) AS fraud_count,
    ROUND(
        SUM(CASE WHEN fraud_label = 1 THEN 1.0 ELSE 0 END)
        / COUNT(*) * 100, 2
    )                                         AS fraud_rate_pct
FROM digital_payment_fraud
GROUP BY transaction_hour
ORDER BY fraud_count DESC
LIMIT 5;
```

Business Answer:

- Fraud peaks around 19:00 (7 PM), 14:00 (2 PM), and 06:00 (6 AM).
- Action: Increase monitoring intensity during these specific hours.

5. Business Recommendations

Based on the SQL analysis findings, the following recommendations are structured by business function.

Priority	Recommendation	Data Signal	Owner
High	Add stronger authentication (OTP / 3D Secure) for Card payments — highest fraud channel	payment_mode, fraud_label	Product
High	Require extra confirmation for Withdrawal and Transfer transactions above a set amount	transaction_type, fraud_label	Risk + Product
Medium	Trigger re-verification once an account crosses 3 previous failed attempts	previous_failed_attempts	Risk Team
Medium	Increase monitoring during the 7 PM, 2 PM, and 6 AM fraud-peak windows	transaction_hour fraud distribution	Operations
Medium	Flag accounts with high login attempts (7+) in 24 hours for extra verification	login_attempts_last_24h	Risk Team
Low	Apply city-level monitoring, starting with Hyderabad and Mumbai (highest fraud volume)	device_location fraud volume	Operations
Low	Don't deprioritize domestic transaction monitoring just because a transaction isn't international	is_international fraud comparison	Risk Team

6. SQL Concepts Demonstrated

SQL Concept	Used In	Purpose
Aggregate Functions (COUNT, SUM, AVG, ROUND)	Q1, Q2, Q3, Q4, Q9, Q12	Summary statistics and fraud metrics

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CASE WHEN (Conditional Logic)	Q1, Q6, Q8, Q11 (all queries)	Flag fraud, create age/risk buckets
GROUP BY	All 12 queries	Segment-level fraud analysis
ORDER BY + LIMIT	Q9, Q12	Sort results and surface top patterns
MIN / MAX	Q2, Q6	Identify amount ranges and bucket boundaries
Type Casting (::NUMERIC)	Q1, Q2	Clean decimal formatting for reporting

7. About This Case Study

Project:

Digital Payment Fraud Detection — SQL Business Case Study

Dataset:

7,500 transactions | 15 features |

Tools & Skills:

PostgreSQL | Aggregate Functions | GROUP BY | CASE WHEN | Business Analytics

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